

Smart Research Collaboration Platform

Master's Project Overview · Mostofa Aminur Rashid · MIT Programme · 2026

1. End-to-End System Flow

① User Registration Profile · Goals · Skills · Papers	② Core Engine Processing Auth · Routing · Indexing	③ AI Matching Algorithm CF+CB · BERT · TrustRank	④ Feature Modules Workspace · Library · Forum	⑤ Publication AssistantJournal Finder · Templates	⑥ Output DeliveryTeam Formed · Paper Ready
---	---	--	---	--	---

INPUT →	PROCESS →	OUTPUT
Profile · Papers · Interests · Skills · Behaviour signals	Matching · Semantic Search · Graph Traversal · Async Queue	Research Team · Structured Project · Published Paper · Profile

2. System Architecture — Layer by Layer

PRESENTATION LAYER	BUSINESS LOGIC LAYER	DATA LAYER
- Next.js (React) - Server-Side Rendering (SSR) - Tailwind CSS UI - Socket.IO real-time sync - Cloudflare CDN + DDoS - Responsive dashboard	- Node.js + Express (API) - JWT / OAuth 2.0 auth - Rate limiting & routingPython - FastAPI (ML) - CF+CB matching engine - Sentence-BERT inference - TrustRank scoring - Kafka async event bus	PostgreSQLUsers · Projects · PostsNeo4j (Graph DB) · Researcher–Paper–Topic graphElasticsearchSemantic + full-text searchRedisPre-cached recommendation feed

Step-by-Step Process

#	Action	What the System Does	Key Technical Detail
1	User Registration	User signs up & fills profile	Domain tags, skills, institution, research goals stored in PostgreSQL
2	Profile Indexing	Profile indexed for matching	Sentence-BERT converts profile text → dense vector embedding
3	Matching Runs	Matching algorithm executes	CF+CB hybrid computes cosine similarity across all profiles
4	Feed Served	Recommendations delivered	Top matches cached in Redis; user sees collaborators, papers, projects
5	Collaboration Begins	User enters collaboration workspace	User joins workspace: task board, doc editor, milestone tracker
6	Publication	Publication Assistant runs	Queries DOAJ/Scimago for journal recommendations

3. Technology Stack Rationale

Layer	Technology	Why This Choice
Frontend	Next.js (React)	SSR = SEO-indexable academic content; faster page load
UI	Tailwind CSS	Utility-first; zero runtime overhead; rapid iteration
Real-time	Socket.IO	WebSocket + auto-fallback; collaborative editing & live notifications
API Gateway	Node.js + Express	Non-blocking I/O; handles 10k+ concurrent connections efficiently
ML Service	Python + FastAPI	ASGI async; native Python ML ecosystem; auto-generates OpenAPI docs
Matching Algo	CF + CBF Hybrid	CBF solves cold-start; CF improves with user behaviour over time
Semantic Search	Sentence-BERT	Dense embeddings understand meaning, not just keywords
Trust Scoring	TrustRank (PageRank)	Credibility flows from verified professors; resists spam/gaming

Primary DB	PostgreSQL	ACID-compliant; handles complex relational queries; battle-tested
Graph DB	Neo4j	Multi-hop traversal (2nd-degree collaborators) in milliseconds
Search Index	Elasticsearch	Unified BM25 + kNN vector search across people, papers, posts
Cache	Redis	Pre-computes recommendation feed as background job; <50ms load
Message Queue	Apache Kafka	Decouples API from ML workers; async processing; durable log
Infrastructure	Docker + Kubernetes	Per-service scaling; ML pods scale independently on load spikes

4. Impact Summary — Problem vs. Solution

#	Problem	ResearchBridge Solution	Measurable Outcome
1	No structured collaborator discovery tool	CF+CB Hybrid + Sentence-BERT surfaces collaborators from profile & behaviour signals	<i>Matched collaborators on first login; relevance improves with use</i>
2	No shared workspace — ad-hoc email/Drive/WhatsApp	Built-in workspace: task board, milestone tracker, real-time doc co-editing	<i>All project activity in one place; version history; no context lost</i>
3	New researchers don't know how to publish	Publication Assistant: journal recommender (DOAJ/Scimago) + templates + checklist	<i>Reduces 'I don't know where to submit' from weeks to minutes</i>
4	Community forums decay into spam	TrustRank algorithm: PageRank-variant weights credibility by citations & endorsements	<i>Quality-first feed; verified institutional badges auto-assigned</i>
5	Cold-start: new users get no useful recommendations	Hybrid approach: CBF bootstraps from explicit tags; CF activates as behaviour accumulates	<i>Even 0-paper researchers get relevant results from day one</i>

Performance Targets

Metric	Current (Manual)	With ResearchBridge	How
Find a collaborator	Weeks (conferences, email, luck)	< 5 minutes	Matching engine on first login
Set up shared workspace	1–2 days (tools, permissions, folders)	< 2 minutes	Auto-created on team formation
Find right journal	Days of research + senior advice	< 10 minutes	Publication Assistant + DOAJ query
Dashboard load time	N/A	< 50 ms	Redis pre-cached feed
Semantic search response	N/A	< 200 ms	Elasticsearch kNN index
Scale to 1M users	Not possible (monolith tools)	Horizontal K8s autoscale	Per-service pod autoscaling

Competitive Gap — What Existing Platforms Are Missing

Feature	Research Gate	Academia. edu	Overleaf	Semantic Scholar	ResearchBridge
Smart collaborator matching	✗	✗	✗	✗	✓ CF+CB+BERT
Collaboration workspace	✗	✗	✓ writing only	✗	✓ Full (tasks, docs, milestones)
Publication assistant	✗	✗	Partial	✗	✓ Journal finder + templates
Mentorship / community	Partial	✗	✗	✗	✓ TrustRank forum
Built for new researchers	✗	✗	✗	✗	✓ Core design goal
Free to deploy (beta)	—	—	—	—	✓ \$0 on free tiers